

#1. Veri setini yükleme ve tanıma

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv("Students Social Media Addiction.csv")

display(df.head())
print(df.shape)
print(df.columns)
```

	Student_ID	Age	Gender	Academic_Level	Country	Avg_Daily_Usag
0	1	19	Female	Undergraduate	Bangladesh	5.2
1	2	22	Male	Graduate	India	2.1
2	3	20	Female	Undergraduate	USA	6.0
3	4	18	Male	High School	UK	3.0
4	5	21	Male	Graduate	Canada	4.5

```
(705, 13)
Index(['Student_ID', 'Age', 'Gender', 'Academic_Level', 'Country',
      'Avg_Daily_Usage_Hours', 'Most_Used_Platform',
      'Affects_Academic_Performance', 'Sleep_Hours_Per_Night',
      'Mental_Health_Score', 'Relationship_Status',
      'Conflicts_Over_Social_Media', 'Addicted_Score'],
      dtype='object')
```

#2.Eksik veri kontrolü

```
print(df.isnull().sum())
```

```
Student_ID      0
Age             0
Gender          0
Academic_Level  0
Country         0
Avg_Daily_Usage_Hours  0
Most_Used_Platform  0
Affects_Academic_Performance  0
Sleep_Hours_Per_Night  0
Mental_Health_Score  0
Relationship_Status  0
Conflicts_Over_Social_Media  0
Addicted_Score  0
dtype: int64
```

#3. Temel istatistiksel analiz

```
display(df.describe())
```

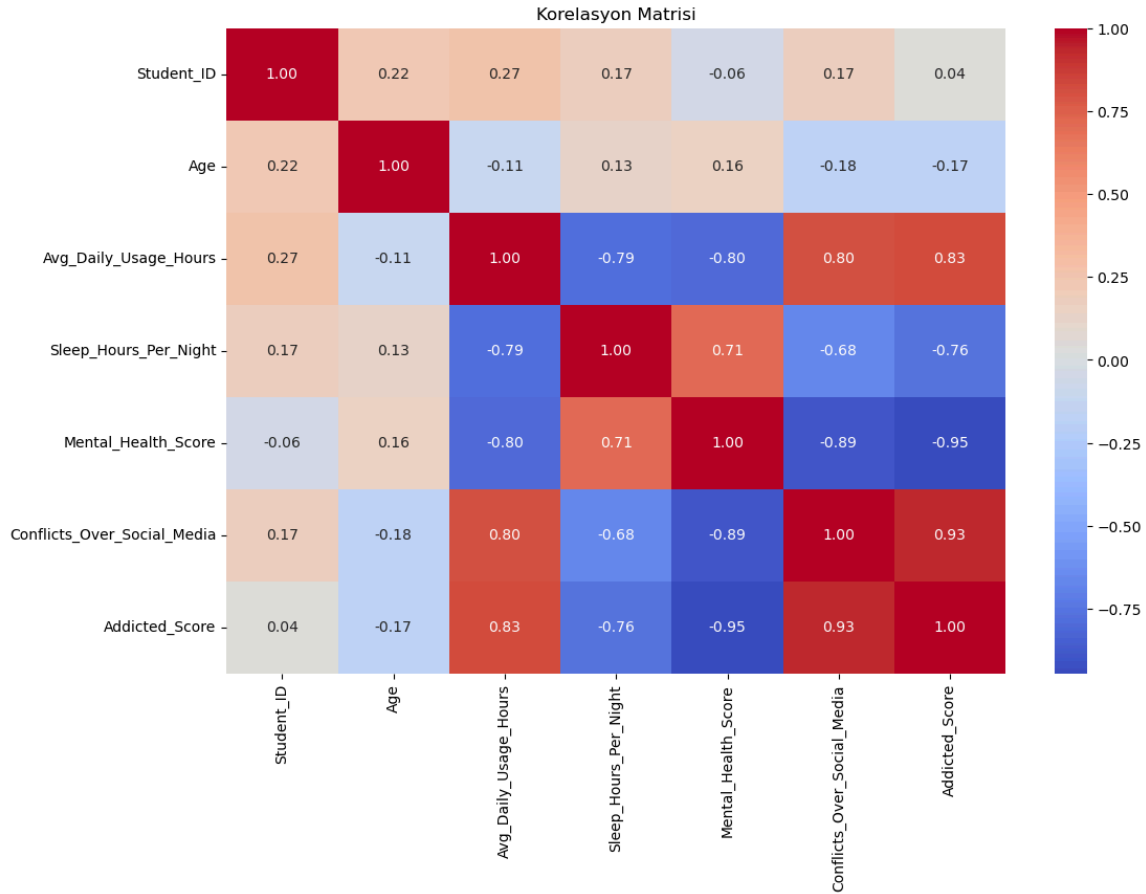
	Student_ID	Age	Avg_Daily_Usage_Hours	Sleep_Hours_Per_Night
count	705.000000	705.000000	705.000000	705.000000
mean	353.000000	20.659574	4.918723	6.868936
std	203.660256	1.399217	1.257395	1.126848
min	1.000000	18.000000	1.500000	3.800000
25%	177.000000	19.000000	4.100000	6.000000
50%	353.000000	21.000000	4.800000	6.900000
75%	529.000000	22.000000	5.800000	7.700000
max	705.000000	24.000000	8.500000	9.600000

#4. Korelasyon analizi

```

numeric_cols = df.select_dtypes(include=["int64", "float64"]).columns
plt.figure(figsize=(12,8))
sns.heatmap(df[numeric_cols].corr(), annot=True, fmt=".2f", cmap="coolwarm")
plt.title("Korelasyon Matrisi")
plt.show()

```



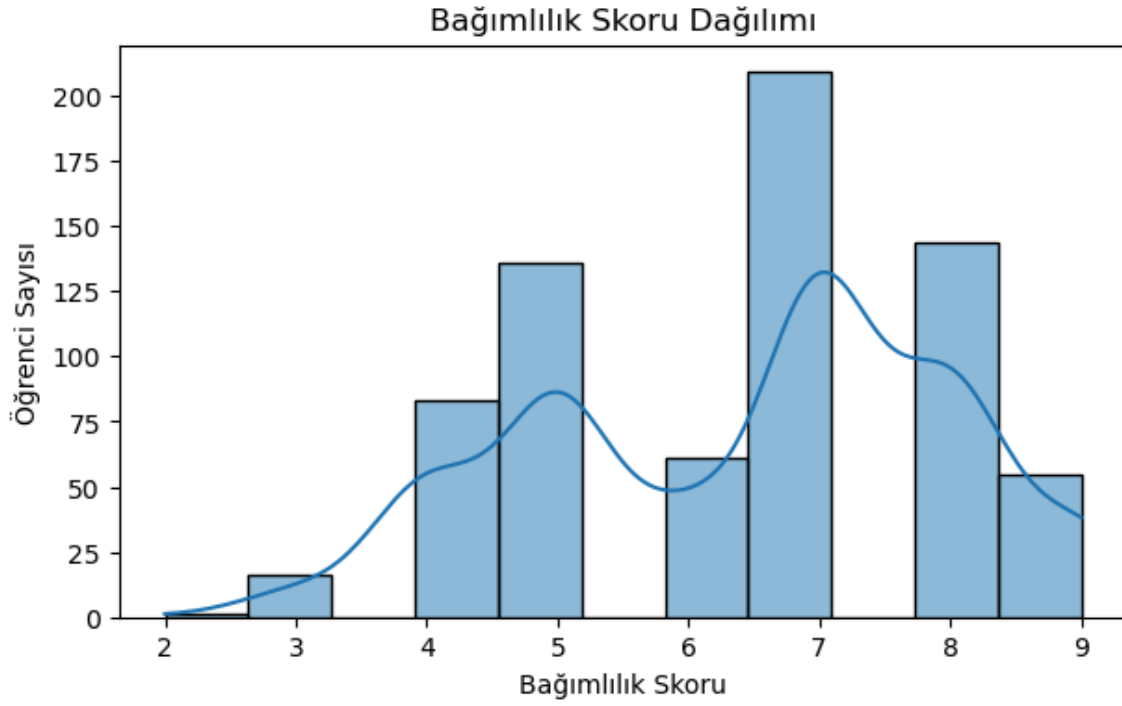
#5. Bağımlılık skoru dağılımı

```

plt.figure(figsize=(7,4))
sns.histplot(df["Addicted_Score"], kde=True)
plt.title("Bağımlılık Skoru Dağılımı")

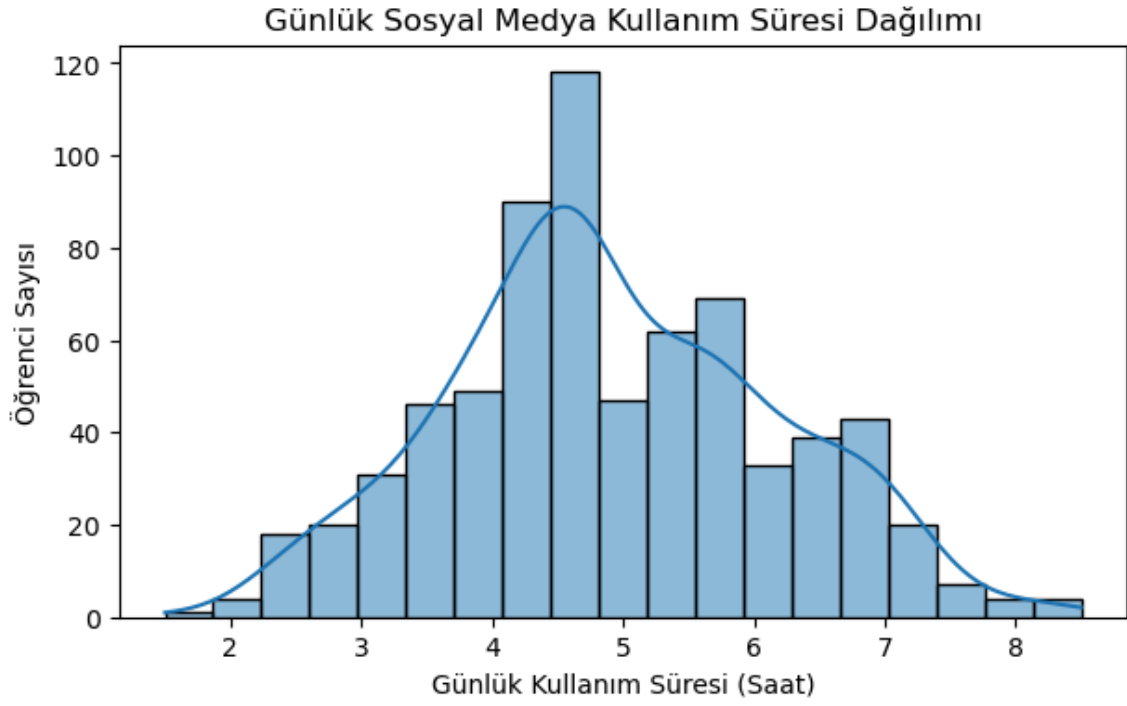
```

```
plt.xlabel("Bağımlılık Skoru")  
plt.ylabel("Öğrenci Sayısı")  
plt.show()
```



#6. Günlük kullanım süresi dağılımı

```
plt.figure(figsize=(7,4))  
sns.histplot(df["Avg_Daily_Usage_Hours"], kde=True)  
plt.title("Günlük Sosyal Medya Kullanım Süresi Dağılımı")  
plt.xlabel("Günlük Kullanım Süresi (Saat)")  
plt.ylabel("Öğrenci Sayısı")  
plt.show()
```



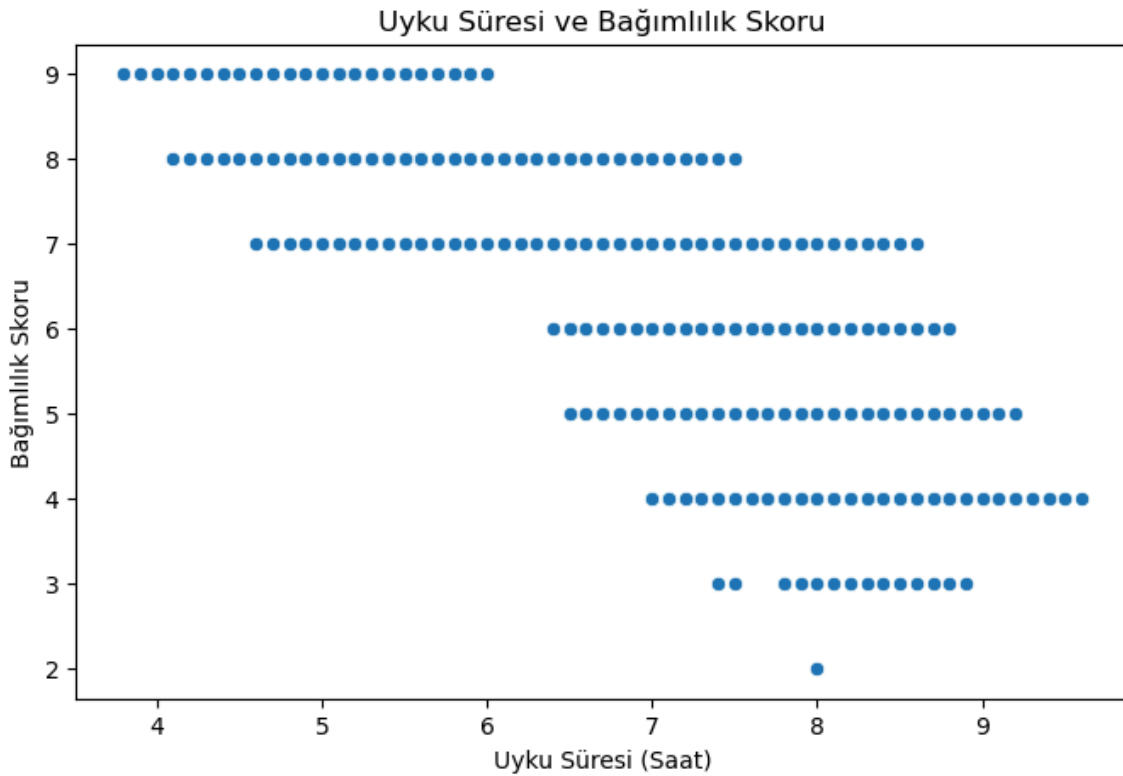
#7. Kullanım süresi ve bağımlılık ilişkisi

```
plt.figure(figsize=(8,5))
sns.regplot(x="Avg_Daily_Usage_Hours", y="Addicted_Score", data=df)
plt.title("Kullanım Süresi ve Bağımlılık Skoru")
plt.xlabel("Günlük Kullanım Süresi (Saat)")
plt.ylabel("Bağımlılık Skoru")
plt.show()
```



#8. Uyku süresi ve bağımlılık ilişkisi

```
plt.figure(figsize=(8,5))
sns.scatterplot(
    x="Sleep_Hours_Per_Night",
    y="Addicted_Score",
    data=df
)
plt.title("Uyku Süresi ve Bağımlılık Skoru")
plt.xlabel("Uyku Süresi (Saat)")
plt.ylabel("Bağımlılık Skoru")
plt.show()
```



#9. Aykırı değer analizi

```
from scipy import stats

z_skor = np.abs(stats.zscore(df["Addicted_Score"]))

print("Z-Score Aykırı Değer İndeksleri:")
print(np.where(z_skor > 3))

Q1 = df["Addicted_Score"].quantile(0.25)
Q3 = df["Addicted_Score"].quantile(0.75)
IQR = Q3 - Q1

alt_sinir = Q1 - 1.5 * IQR
ust_sinir = Q3 + 1.5 * IQR

aykiri_degerler = df[
```

```

(df["Addicted_Score"] < alt_sinir) |
(df["Addicted_Score"] > ust_sinir)
]

```

```
display(aykiri_degerler)
```

Z-Score Aykırı Değer İndeksleri:
(array([], dtype=int64),)

	Student_ID	Age	Gender	Academic_Level	Country	Avg_Daily_Usage_H
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#10. Basit doğrusal regresyon

```
import statsmodels.api as sm
```

```

Y = df["Addicted_Score"]
X = df[["Avg_Daily_Usage_Hours"]]
X = sm.add_constant(X)

```

```
model = sm.OLS(Y, X).fit()
```

```
print(model.summary())
```

```

                                OLS Regression Results
=====
Dep. Variable:                  Addicted_Score    R-squared:
0.692
Model:                            OLS          Adj. R-squared:
0.692
Method:                         Least Squares   F-statistic:
1581.
Date:                            Sat, 30 May 2026   Prob (F-statistic):
182                                4.70e-
Time:                            22:55:58         Log-Likelihood:
-910.15
No. Observations:                705             AIC:
1824.
Df Residuals:                    703             BIC:
1833.
Df Model:                        1
Covariance Type:                nonrobust
=====
                                coef    std err          t      P>|t|      [0.025
0.975]
-----
const                1.2712     0.134     9.481     0.000     1.008
1.534
Avg_Daily_Usage_Hours  1.0502     0.026    39.763     0.000     0.998
1.102
=====
Omnibus:                    9.870   Durbin-Watson:
1.819

```

Prob(Omnibus): 0.007 Jarque-Bera (JB):
6.568
Skew: 0.080 Prob(JB):
0.0375
Kurtosis: 2.555 Cond. No.
21.3

=====

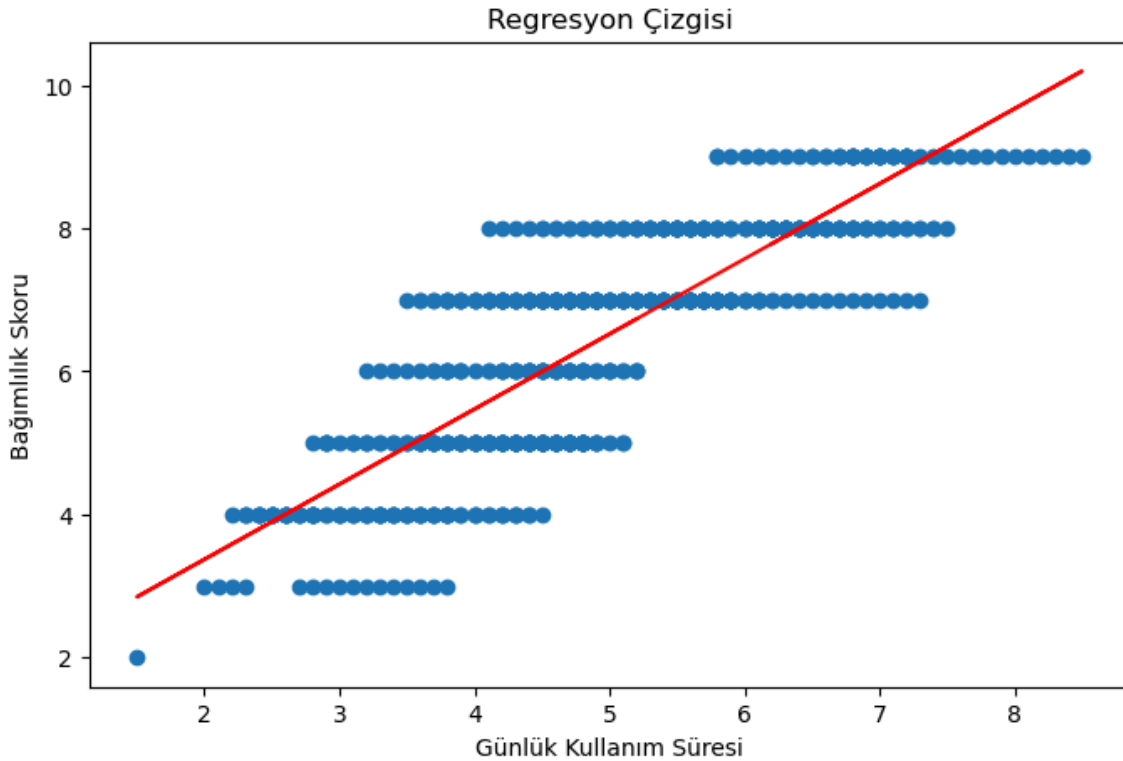
Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

#11. Regresyon çizgisi

```
tahmin = model.predict(X)
```

```
plt.figure(figsize=(8,5))  
plt.scatter(df["Avg_Daily_Usage_Hours"], df["Addicted_Score"])  
plt.plot(df["Avg_Daily_Usage_Hours"], tahmin, color="red")  
plt.xlabel("Günlük Kullanım Süresi")  
plt.ylabel("Bağımlılık Skoru")  
plt.title("Regresyon Çizgisi")  
plt.show()
```



#12. Çoklu regresyon

```
Y2 = df["Addicted_Score"]
```

```
X2 = df[["Avg_Daily_Usage_Hours",  
         "Sleep_Hours_Per_Night",
```

```

    "Mental_Health_Score",
    "Conflicts_Over_Social_Media"
]]

```

```

X2 = sm.add_constant(X2)

```

```

coklu_model = sm.OLS(Y2, X2).fit()

```

```

print(coklu_model.summary())

```

```

                                OLS Regression Results
=====
Dep. Variable:                  Addicted_Score    R-squared:
0.944
Model:                            OLS    Adj. R-squared:
0.944
Method:                        Least Squares    F-statistic:
2978.
Date:                        Sat, 30 May 2026    Prob (F-statistic):
0.00
Time:                        22:56:15    Log-Likelihood:
-306.32
No. Observations:                  705    AIC:
622.6
Df Residuals:                      700    BIC:
645.4
Df Model:                          4
Covariance Type:                  nonrobust
=====

                                coef    std err          t      P>|t|
-----
[0.025    0.975]
-----
const                            10.0619      0.336     29.979      0.000
9.403    10.721
Avg_Daily_Usage_Hours            0.0209      0.023      0.901      0.368
-0.025     0.067
Sleep_Hours_Per_Night           -0.2100      0.021    -10.032      0.000
-0.251    -0.169
Mental_Health_Score             -0.6715      0.030    -22.085      0.000
-0.731    -0.612
Conflicts_Over_Social_Media      0.6652      0.035     19.090      0.000
0.597     0.734
=====
Omnibus:                        30.736    Durbin-Watson:
2.056
Prob(Omnibus):                  0.000    Jarque-Bera (JB):
41.335
Skew:                          0.402    Prob(JB):
09                                1.06e-
Kurtosis:                      3.872    Cond. No.
261.
=====

```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.